

Did you know?

- Rubber Duck Debugging or Rubber Ducking is the practice of explaining your code, line by line to a rubber duck, so that you come across the mistakes by yourself.
- A Heisenbug is a bug that disappears when you try to debug it using tools like *printf*, and comes back once you restore the code. Its name is a pun on the name of Heisenberg, who introduced the Uncertainty Principle.

Breakpoints in GDB

To start with, you can set breakpoints in the GDB shell using *b file:line* or *b functionname*. Now you can run the program as usual. Once the program reaches the specified lines, it pauses and lets you perform various status checks. You can continue using the *continue (c)* command or delete the breakpoint at that line using the *clear command*, once you are done. *clear file:line* will delete the breakpoint in a specific line while *clear functionname* will delete all the breakpoints under it.

The *delete breakpointnumber* command (shorthand: *d*) will delete the specified breakpoint while the same command with no breakpoint number will delete all the breakpoints with a confirmation.

The very next step with GDB is perhaps Stepping, which lets you step through your program, line by line, without setting a thousand breakpoints. But this is out of the scope of this article, and we are leaving GDB here.

Let's now list some utilities that will help you deal with programs written in C and much lower-order languages:

- *strace* — to trace system calls and signals
- *objdump* — to display information from object files, including disassembly
- *mtrace* — to find memory leaks
- *hexdump* — ASCII, decimal, hexadecimal, octal dump, especially to analyse binary files
- Static code analysis tools and visualisers **END**

References

- [1] <https://lcmire.me/blog/2016/06/21/i-do-not-use-a-debugger/>
- [2] <https://www.gnu.org/software/gdb/documentation/>

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